

习题课: 群(Group)

4.14

1. 代数结构 (群, 环, 域)
集合 + 运算 $(\mathbb{N}, +)$
 $(\mathbb{N}, \times)$
 $\mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$
 $\text{Mat}_{n \times n}(\mathbb{R})$ $(t, \times)$
无支撑律

2. 半群, 么半群 (monoid), 群, Abel 群

半群: ① 运算: 封闭 = 元 (1个操作数)

② 结合律 $a \cdot (bc) = (a \cdot b) \cdot c$

么半群: ③ 么元 / 单位元: $e / 1$ $e \cdot a = a \cdot e = a$

群: ④ 逆元 a^{-1} $a^{-1} \cdot a = a \cdot a^{-1} = e$

Exercise

Let S be a set and define a product in S by $ab = b$. Show that S is a semigroup. Under what condition does S contain a unit?

① $\cdot$: closure + binary

② associativity

1. ① $\forall a, b \in S, a \cdot b = b \in S$

② $(a \cdot b) \cdot c = b \cdot c = c. \implies a \cdot (b \cdot c) = a \cdot c = c.$

2. $\forall a, \begin{cases} e \cdot a = a \\ a \cdot e = a \end{cases} S = \{e\}.$
 $\hookrightarrow e$

Let $M = \mathbb{Z} \times \mathbb{Z}$ the set of pairs of integers (x_1, x_2) . Define $(x_1, x_2)(y_1, y_2) = (x_1 y_1 + 2x_2 y_2, x_1 y_2 + x_2 y_1)$, $1 = (1, 0)$. Show that this defines a monoid. (Observe that the commutative law of multiplication holds) Show that if $(x_1, x_2) \neq (0, 0)$ then the cancellation law will hold for (x_1, x_2) , that is, $(x_1, x_2)(y_1, y_2) = (x_1, x_2)(z_1, z_2) \Rightarrow (y_1, y_2) = (z_1, z_2)$

1. ① closure + binary ④ unit

② associativity

群有消去律, 但有左右消去律也不一定是群.

$$2. (x_1, x_2)(y_1, y_2) = (x_1, x_2)(z_1, z_2)$$

$$\begin{vmatrix} x_1 & 2x_2 \\ x_2 & x_1 \end{vmatrix} = x_1^2 - 2x_2^2 \neq 0$$

Let $(M, p, 1)$ be a monoid and let $m \in M$. Define a new product p_m in M by $p_m(a, b) = amb$. Show that this defines a semigroup. Under what condition on m do we have a unit relative to p_m ?

3. 证明给定集合+运算是个代数结构 / 判断

4. 什么是群?

2. 单边定义 $\checkmark$ (左逆元+左逆元 $\Rightarrow$ 群, 右+右 $\checkmark$ 左+右 $\times$ 右+左 $\times$)

③ 左逆元: $e \cdot a = a \checkmark$ $a \cdot e \neq a$ 设 $a_1 = a^{-1}$ (左) $a_2 = (a^{-1})^{-1}$

④ 左逆元: $a^{-1} \cdot a = e$ $a \cdot a^{-1} \neq e$ 即 $a_1 \cdot a = e$, $a_2 \cdot a_1 = e$

3. 方程定义 $a \cdot a_1 = e(a \cdot a_1) = (a_2 a_1)(a a_1) = a_2(a_1 a) a_1$

性质: G 是群, $\forall a, b \in G$. $= a_2(e a_1) = a_2 a_1 = e$. a_1 是右逆元

$$a \cdot x = b \quad / \quad x \cdot b' = a'$$

$$a^{-1} \cdot a \cdot x = a^{-1} \cdot b \quad x = a^{-1} \cdot b \quad \text{可解}$$

半群 S , $\forall a, b \in G$. $ax=b$. $xa=b$ 都有解. $\Rightarrow S$ 为群

① identity(unit) $\forall a \in G$, $xa=a$ 记解 $x=e_a$ (a 的左单位元)

$\forall c \in G$, $ax=c$. 记解 $x=d$, $ad=c$

$e_a \cdot c = e_a \cdot ad = ad = c$. e_a 也是 c 的左单位元. $\therefore e_a$ 是所有元素的左单位元.

④ $xb=e_a$ 有解 记 $x=d$ $db=e_a$ d 是 b 的左逆元.

根据单边定义, 是群.

Basic Algebra

1.2 T11

$(\mathbb{Z}^+ \cup \{0\}, +)$
 $3+5=3+5$ 消去律
 -3 不在 $\mathbb{Z}^+ \cup \{0\}$ 中.

函数复合, 置换, 矩阵乘法
 不满足交换律

4. 有限半群 满足消去律, 为群

消去律

$S = \{a_1, a_2, \dots, a_n\}$ $a_i \cdot S = \{a_i a_1, a_i a_2, \dots, a_i a_n\}$ 不重复 (由消去律)

$a_i S \subseteq S$, $a_i a_j \in S$. (半群运算封闭) $|a_i S| = |S|$. 故 $a_i S = S$

$\forall a_i, a_j, a_k$ $a_i a_j = a_k$ 有解. (用了定义3)

$a_i S = S$

5. 子群 $\rightarrow$ 陪集 $\rightarrow$ 拉格朗日 $|H| \mid |G|$

$\downarrow \rightarrow$ 正规子群, 商群 (除法)

循环群 (生成群) $\langle a \rangle = \{na \mid n \in \mathbb{Z}\} \leq G$

$\langle S \rangle$ $S = \{a, b, c\}$

6. 群同态 / 同构 $\langle \{a, b\} \rangle = \{ma + nb \mid n, m \in \mathbb{Z}\}$

$\phi: G_1 \rightarrow G_2$

$a \mapsto \phi(a)$

$\phi(a+b) \mapsto \phi(a)\phi(b)$

$(\mathbb{Z}_n, \oplus_n)$ $(\mathbb{Z}, +)$ 循环群, 同构

7. Sylow P 80, 81 Basic Algebra

EXERCISES

先算 β $\star$ 先算 α $\alpha_1 \circ \alpha_2 \circ \dots \circ \alpha_n$
 1 Determine $\alpha\beta$, $\beta\alpha$ and α^{-1} in S_5 if

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{pmatrix}$$

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$$

$$\alpha^{-1} = \begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$$

- 4 Let G be the set of pairs of real numbers (a, b) with $a \neq 0$ and define $(a, b)(c, d) = (ac, ad + b)$, $1 = (1, 0)$. Verify that this defines a group.

① binary + closure $\rightarrow +$.

② associativity $((a, b)(c, d))(e, f) = (a, b)((c, d)(e, f))$

③ identity: $1 = (1, 0)$

④ reverses: $\begin{cases} ac = 1 \\ ad + b = 0 \end{cases} \Rightarrow \begin{cases} c = \frac{1}{a} \\ d = -\frac{b}{a} \end{cases} \quad \left(\frac{1}{a}, -\frac{b}{a}\right)$

5. Let G be the set of transformations of the real line $\mathbb{R}$ defined by $x \rightarrow x' = ax + b$ where a and b are real numbers and $a \neq 0$. Verify that G is a transformation group of $\mathbb{R}$.

变换的群
(映射)

$$\phi_{a,b} \quad x \mapsto ax + b$$

$$\phi_{c,d} \quad x \mapsto cx + d$$

$$\phi_{ab} \circ \phi_{cd} = a(cx + d) + b = \textcircled{ac}x + \textcircled{ad + b} \quad \text{和 T4 同.}$$

13. Show that any finite group of even order contains an element $a \neq 1$ such that $a^2 = 1$.

偶

$$a \neq 1, a^2 = 1 \Rightarrow a = a^{-1}$$

置换的阶

$$(1, 2, 7) \in S_7 \quad \text{阶为 3}$$

$$\text{阶 2, 阶 4} \quad \text{lcm}(2, 4) = 4.$$

$$(1, 2, 7)(3, 4, 5, 6) \quad \text{阶为 12}$$

3

4

$$\text{lcm}(3, 4) = 12$$

定义. $(R, +, \cdot)$ 满足

(1) $(R, +)$ 交换群

(2) $(R, \cdot)$ 是 $(\underline{})$ 半群

(3) $\forall a, b, c \in R, a(b+c) = ab+ac, (a+b)c = ac+bc$

若 $(R, \cdot)$ 有交换律, 即 $\forall a, b \in R, ab=ba$, 则称 $(R, +, \cdot)$ 是交换环

$(\mathbb{Z}, +, \cdot) \quad (M_2(\mathbb{R}), +, \cdot)$

eg1. $(\mathbb{Z}_2 = \{0, 1\}, \oplus, \cdot)$

模2加

$0 \oplus 0 = 0, 0 \oplus 1 = 1 \oplus 0 = 1, 1 \oplus 1 = 0$ 封闭, 交换

$(\mathbb{Z}_2, \oplus)$ 是交换群, $\oplus$ 下么元为 0 $(\mathbb{Z}_2, \cdot)$ 是么半群, $\cdot$ 下么元为 1

$a(b \oplus c) = ab \oplus ac$ 分配律

若 $a=0$, 左 = 右 = 0

$a=1$, 左 = 右 = $b \oplus c$

$(b \oplus c)a = ba \oplus ca$

eg2. 设 n 是正整数, $\mathbb{Z}_2^n = \{0, 1\}^n = \{(a_1, \dots, a_n) \mid \text{每 } a_i \in \{0, 1\}\}$

$(\mathbb{Z}_2^n, \oplus, \cdot)$

模2加

$(a_1, \dots, a_n) \oplus (b_1, \dots, b_n) = (a_1 \oplus b_1, a_2 \oplus b_2, \dots, a_n \oplus b_n)$

$(a_1, \dots, a_n)(b_1, \dots, b_n) = (a_1 b_1, a_2 b_2, \dots, a_n b_n)$

$(\mathbb{Z}_2^n, \oplus)$ 是交换群, $\oplus$ 下么元 $(0, 0, \dots, 0)$

$(\mathbb{Z}_2^n, \cdot)$ 是么半群, $\cdot$ 下么元 $(1, 1, \dots, 1)$

分配律

对称差



对称差 $A \oplus B = (A - B) \cup (B - A)$
 $= (A \cup B) - (A \cap B)$



eg3. 给定集合 X , $P(X) = \{A \mid A \subseteq X\}$ ($P(X)$, $\oplus$, $\cap$) 是环
 $(P(X), \oplus)$ 是交换群, $\oplus$ 下元 ϕ , $A \oplus \phi = \phi \oplus A = \phi$ $A \oplus A = \phi$
 $(A \oplus B) \oplus C = (A - (B \cup C)) \cup (B - (A \cup C)) \cup (C - (A \cup B)) \cup (A \cap B \cap C)$
 $(P(X), \cap)$ 是么半群, $\cap$ 下元 X , $A \cap X = X \cap A = A$
分配律 $A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$

环同构

$(\mathbb{Z}_2^n, \oplus, \cdot)$ 与 $(P(X), \oplus, \cap)$ 的联系.

$$X = \{1, 2, \dots, n\}$$

$$(a_1, a_2, \dots, a_n) \mapsto A = \{i \mid 1 \leq i \leq n, a_i = 1\} \quad (\text{支撑集})$$

$\cap$
 $\mathbb{Z}_2^n$

$\cap$
 $P(X)$

$$\varphi: \mathbb{Z}_2^n \longrightarrow P(X)$$

$$\varphi(a_1, a_2, \dots, a_n) = \{i \mid 1 \leq i \leq n, a_i = 1\} \quad \text{可证明是双射, 个数: } 2^n$$

单射: 若 $\varphi(a_1, \dots, a_n) = \varphi(b_1, \dots, b_n)$, 则 $a_i = b_i$.

满射: $\forall A \subseteq X$, 在 A 中的 i , a_i 设为 1, 其余 0. 故都能找到 $\mathbb{Z}_2^n$ 中数.

验证运算联系

加法群同构

$$\alpha = (a_1, \dots, a_n) \quad \beta = (b_1, \dots, b_n)$$

$$\varphi(\alpha) = A \quad \varphi(\beta) = B$$

$$\varphi(\alpha \oplus \beta) = A \oplus B$$

$$\alpha \oplus \beta = (a_1 \oplus b_1, a_2 \oplus b_2, \dots, a_n \oplus b_n)$$

$$a_i \oplus b_i = 1 \Leftrightarrow (a_i = 0, b_i = 1) \text{ 或 } (a_i = 1, b_i = 0)$$

$$\Leftrightarrow (i \in B - A) \text{ 或 } (i \in A - B)$$

$$\Leftrightarrow i \in A \oplus B.$$

$$\varphi(\alpha \beta) = A \cap B$$

$$(a_1, \dots, a_n)(b_1, \dots, b_n) = (a_1 b_1, \dots, a_n b_n)$$

$$a_i b_i = 1 \Leftrightarrow a_i = b_i = 1 \Leftrightarrow i \in A \cap B.$$

eg4.

群环

G 是有限群, $\mathbb{C}$ 复数

$$\mathbb{C}^G = \{f: G \rightarrow \mathbb{C}\}$$

$$(\mathbb{C}^G, +, *)$$

$$f, g \in \mathbb{C}^G$$

$$(f+g)(a) = f(a) + g(a) \quad (\forall a \in G) \quad // \text{分量相加}$$

$$(f * g)(a) = \sum_{b \in G} f(b) g(b^{-1}a) = \sum_{\substack{(b,c) \in G \times G \\ bc=a}} f(b) g(c)$$

// 群有消去律 $\Rightarrow c = b^{-1}a$

$$(\mathbb{C}^G, +) \text{ 交换群}$$

$$(\mathbb{C}^G, *) \text{ 么半群 么元?} \quad // \text{由群结合律, 群么元}$$

分配律 // 由复数运算分配律

eg5.

四元数环

$$R = \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C} \right\} \quad // \alpha = a + b\sqrt{-1}, a, b \in \mathbb{R}$$

$$\bar{\alpha} = a - b\sqrt{-1} \quad |\alpha| = \sqrt{a^2 + b^2}$$

验证封闭性 // 则分配律, 结合律继承

$$\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} + \begin{pmatrix} \gamma & \theta \\ -\bar{\theta} & \bar{\gamma} \end{pmatrix} = \begin{pmatrix} \alpha + \gamma & \beta + \theta \\ -(\bar{\beta} + \bar{\theta}) & \bar{\alpha} + \bar{\gamma} \end{pmatrix}$$

$$\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \begin{pmatrix} \gamma & \theta \\ -\bar{\theta} & \bar{\gamma} \end{pmatrix} = \begin{pmatrix} \alpha\gamma - \beta\bar{\theta} & \alpha\theta + \beta\bar{\gamma} \\ -\bar{\beta}\gamma - \bar{\alpha}\bar{\theta} & -\bar{\beta}\theta + \bar{\alpha}\bar{\gamma} \end{pmatrix} = \begin{pmatrix} \alpha\gamma - \beta\bar{\theta} & \alpha\theta + \beta\bar{\gamma} \\ -(\alpha\bar{\theta} + \beta\bar{\gamma}) & \alpha\bar{\gamma} - \beta\bar{\theta} \end{pmatrix}$$

☆☆

eg6.

模 n 的剩余类环 $n \in \mathbb{Z}^+, n \geq 2$

$$(\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}, \oplus, \otimes)$$

模 n 加 模 n 乘

$$a \oplus b = (a+b) \% n \quad a \otimes b = (ab) \% n$$

$$a \oplus b \equiv a+b \pmod{n} \quad a \otimes b \equiv ab \pmod{n}$$

$(\mathbb{Z}_n, \oplus)$ 循环群, $\oplus$ 下幺元 0

$(\mathbb{Z}_n, \otimes)$ 交换么半群, $\otimes$ 下幺元 1

分配律 运算方便, 都在 $0 \sim n-1$ 内

$$a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$$

写起来方便 $\leftarrow (ab+ac) \% n$

$$a \otimes (b \oplus c) \equiv a(b+c) \pmod{n}$$

$$a \otimes b = (ab) \% n = (ba) \% n = b \otimes a$$

$$a \otimes 1 = (a \cdot 1) \% n = a \% n = a$$

$$(a \otimes b) \otimes c = a \otimes (b \otimes c) = (abc) \% n$$

$$(a \otimes b) \otimes c \equiv (a \otimes b) c \pmod{n}$$

$$a \otimes b \equiv ab \pmod{n} \quad (a \otimes b) c \equiv abc \pmod{n}$$

$$(a \otimes b) \otimes c \equiv abc \pmod{n} \quad (a \otimes b) \otimes c = (abc) \% n$$

定理. 设 $(R, +, \cdot)$ 是环, $a, b \in R$, 则

$$(1) a0 = 0a = 0$$

// 0: 加法幺元

$$(2) -(ab) = (-a)b = a(-b)$$

// $-a$: 加法逆元

证: (1) $a0 = a(0+0) = a0 + a0$ (分配律)

由 $(R, +)$ 的消去律, $0 = a0$

$$0a = (0+0)a = 0a + 0a \text{ (分配律)}$$

由 $(R, +)$ 的消去律, $0 = 0a$

$$(2) 0 = a0 = a(b+(-b)) = ab + a(-b)$$

$$a(-b) = -(ab)$$

$$0 = 0b = (a+(-a))b = ab + (-a)b$$

$$(-a)b = -(ab)$$